Genus-3 GHS covers of binary curves over F_{2^{3l}}: census and

subgroup shape Study `ghs_n3_genus3_census_20261024`, run 2026-10-24. Protocol: `PROTOCOL.md` (frozen before data; sha256 `7401a1de746a6b516785a5f370aa286c29bf5d4d8f5195d8124f093d2b87b642`, unchanged after the run). Tool: `tool/census.rs`. Raw output: `data/`. Every number is labelled **derived** (from the algebra in PROTOCOL.md section 1) or **measured** (computed by the tool). Toy fields only (N = 3l <= 21); no claim about any deployed curve, no DLP cost or speed claim.

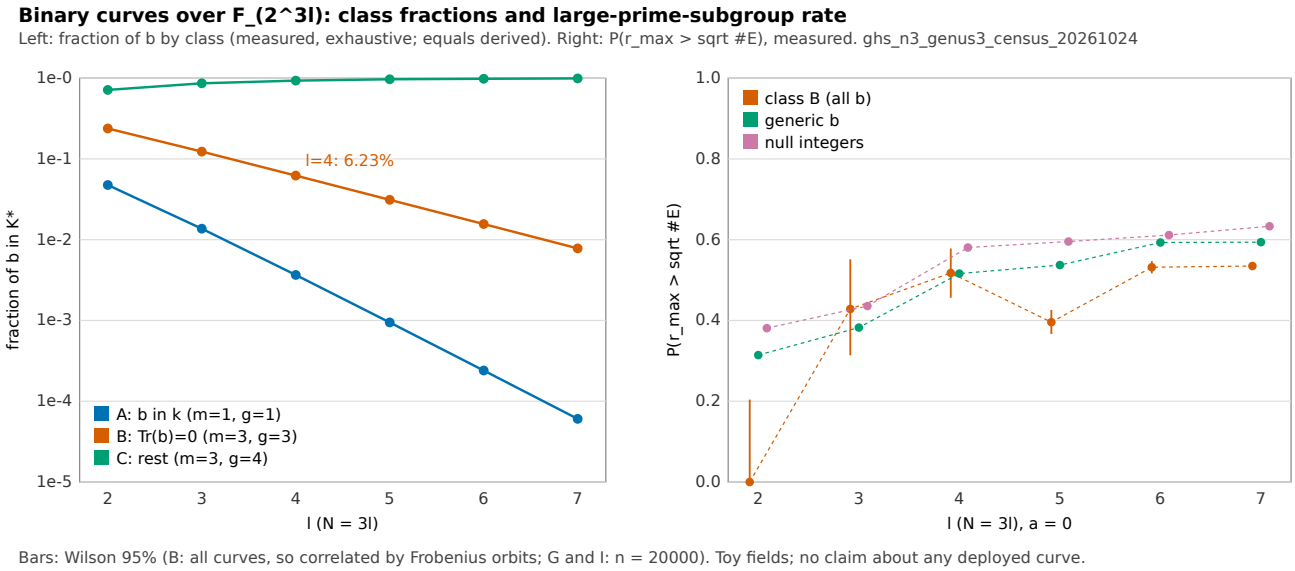


Figure 1: class fractions and large-prime rate

Figure. Left: fraction of b in K by GHS class vs l (measured, exhaustive; the derived closed forms coincide with every point). Right: P(largest prime factor of #E > sqrt #E) for a = 0, class B vs generic b vs null integers (measured; Wilson 95%). Source data: `data/q1_census.csv`, `data/q3_shaped.csv`; run log `data/run.log`.*

Reproduction

```
cd /home/user/crypto   # target/release built beforehand (library + ghs_screen)
rustc --edition 2021 -O research/ghs_n3_genus3_census_20261024/tool/census.rs \
  -o /tmp/census --extern crypto_lib=target/release/deps/libcrypto_lib-2ea5067956474c6e.rlib \
  --extern num_bigint=target/release/deps/libnum_bigint-72cf526ce00d5673.rlib \
  -L dependency=target/release/deps
/tmp/census research/ghs_n3_genus3_census_20261024/data target/release/ghs_screen
```

Single thread, 2 min 26 s wall on a shared 4-core host. No file in `src/`, `examples/` or `Cargo.toml` was touched. The library calls are `ghs_genus_with_type`, `magic_number`, `mmt_condition_8`, `relative_trace`, `absolute_trace`. The tool reimplements only exp/log tables, the point count, factoring, Wilson intervals and the SVG.

Q1. Closed form for the (m, type, genus) census (derived, verified)

For n = 3 the Frobenius sigma = 2^l-power has order 3, and x^3 - 1 = (x+1)(x^2+x+1) splits K = k (+) ker Tr_{K/k} as F_2[sigma]-modules (dimensions l and 2l). The annihilator of sqrt(b) is the product of (x+1) if Tr_{K/k}(b) != 0 and (x^2+x+1) if sqrt(b) has a nonzero trace-zero part. So only m in {1, 3} occurs (never m = 2 or m >= 4):

class	condition on b	m	type	genus	count (derived)
A	b in F_{2^l}*	1	II	1	2^l - 1
B	Tr_{K/k}(b) = 0, b != 0	3	I	**3**	2^{2l} - 1
C	otherwise	3	II	4	(2^l - 1)(2^{2l} - 1)

Measured (exhaustive over all 2^N - 1 values of b, library classification):

l	N	A obs/derived	B obs/derived	C obs/derived	B fraction	per-b deviations
2	6	3 / 3	15 / 15	45 / 45	23.8%	0
3	9	7 / 7	63 / 63	441 / 441	12.3%	0
4	12	15 / 15	255 / 255	3825 / 3825	6.23%	0
5	15	31 / 31	1023 / 1023	31713 / 31713	3.12%	0
6	18	63 / 63	4095 / 4095	257985 / 257985	1.56%	0

7	21	127 / 127	16383 / 16383	2080641 / 2080641	0.78%	0

P1 held: zero deviations for every b; the library's class equals the predicate (b in k) / (Tr_{K/k} b = 0) for each of the 2^N - 1 values at every l. The l = 4 census (15 / 255 / 3825) is reproduced. The 6% therefore falls like 2^{-l}: by l = 7 only 0.78% of b are genus 3.

Q2. MMT condition (8) (derived; measured to agree)

Condition (8) is (n odd) or Tr(a)=0 or Tr_{K/k}(b) != 0; n = 3 is odd. **P2 held**: 100% of genus-3 b satisfy it for a = 0 and for Tr_{K/F2}(a) = 1, at every l = 2..7 (`data/q2_cond8.csv`; also 100% in classes A and C). For n = 3 the descent condition is vacuous, so this is not a sieve.

Detector positive control (n = 2, derived expectation (2^{2l} - 2^l) of 2^{2l} - 1 for Tr(a)=1, 100% for a=0): measured l=3: 56/63, l=4: 240/255, l=5: 992/1023, a=0: all true. All three equal the derived counts, so the tally can see a failing condition when one exists (P2c held).

Q3. Do the class-B curves have a cryptographically shaped subgroup?

Deviation from the brief (see PROTOCOL.md section 3). "r does not divide #E(F_{2^l})" is undefined for class B: b is not in k, so E has no model over k. Evaluated instead (measured): S1 = largest prime r_max of #E(K) exceeds sqrt(#E); S2 = cofactor #E/r_max <= 4. The literal criterion is evaluated only for b in k (arm D, where E is defined over k): there it coincides with S1 in every row (`data/run.log`, "Arm D"), because #E(k) <= q + 1 + 2 sqrt q < q^{3/2} < sqrt #E(K) (derived), so no prime above sqrt #E(K) can divide #E(k). The trace constraint is never what removes a large prime; the question is only the shape of #E(K) itself.

Point counts: exhaustive over x (trace formula, PROTOCOL.md section 3), checked by a brute-force affine-point count at l = 2, 3 (every b, both a: 0 mismatches), and for every class-B curve at every l by Hasse bound (0 violations), #E even, #E mod 4 equal to 0 (a=0) / 2 (Tr a=1) (0 violations), twist sum 2*2^N+2 (0), and Frobenius-orbit count equality (0).

S1 and S2, measured (a = 0 and Tr(a)=1). B is over all class-B b; "B reps" is over one b per Frobenius orbit (the independent sample, Wilson 95%); G is 20000 uniformly random b; I is 20000 integers in the Hasse interval with the same #E mod 4.

l	a	S1 B (n)	S1 B reps [95%]	S1 G	S1 I	S2 B	S2 G	S2 I
4	a=0	0.518 (255)	0.478 [0.292,0.670]	0.516	0.581	0.000	0.151	0.157
4	Tr=1	0.647 (255)	0.652 [0.449,0.812]	0.695	0.727	0.377	0.260	0.301
5	a=0	0.396 (1023)	0.391 [0.285,0.509]	0.537	0.595	0.000	0.070	0.104
5	Tr=1	0.575 (1023)	0.580 [0.462,0.689]	0.678	0.741	0.147	0.130	0.185
6	a=0	0.532 (4095)	0.524 [0.460,0.587]	0.593	0.611	0.000	0.087	0.095
6	Tr=1	0.756 (4095)	0.758 [0.698,0.808]	0.712	0.738	0.149	0.128	0.159
7	a=0	0.535 (16383)	0.535 [0.500,0.570]	0.594	0.633	0.000	0.049	0.075
7	Tr=1	0.705 (16383)	0.706 [0.673,0.736]	0.697	0.727	0.094	0.103	0.148

(l = 2, 3 are in `data/q3_shaped.csv`; their samples are 15 and 63 curves.)

Number of class-B curves with a shaped subgroup (measured, S1, all b, not orbit-reduced; the curves come in Frobenius orbits of size up to 3l): a = 0: 132 / 405 / 2178 / 8760 and Tr(a)=1: 165 / 588 / 3096 / 11553 for l = 4 / 5 / 6 / 7. As a fraction of all 2^N - 1 curves (derived from these counts): a = 0: 132/4095 = 3.2%, 405/32767 = 1.2%, 2178/262143 = 0.83%, 8760/2097151 = 0.42%; Tr(a)=1: 165/4095 = 4.0%, 588/32767 = 1.8%, 3096/262143 = 1.2%, 11553/2097151 = 0.55%. Of the genus-3 share itself, the shaped part is 0.40-0.76 (table). "The 6% survives" therefore only in the sense that about half to three quarters of the genus-3 curves are shaped, and the 6% itself halves with each increment of l.

P3 (class B indistinguishable from generic and null) was partly rejected. Pooled two-proportion z of B reps vs G (`data/zscores.txt`, measured): for Tr(a)=1, |z| <= 1.75 at l = 4..7 (consistent with generic; vs the null integers l = 5 gives z = −3.06, the other l are within 1.4). For a = 0, B is lower than generic at l = 5, 6, 7 (z = −2.43, −2.13, −3.27) and than the null integers (z = −3.45, −2.71, −5.58); l = 4 is not significant (n = 23 orbits). Unplanned (post hoc, measured, no derivation yet) observation that accompanies it: for a = 0 every class-B curve has 8 | #E(K) (minimum v2(#E) = 3 at every l = 3..7, 63 to 16383 curves), where a generic a = 0 curve only has 4 | #E (v2 >= 2). That forced extra 2-power also explains why S2 (cofactor <= 4) is exactly 0 for a = 0 class-B curves. For Tr(a)=1, v2(#E) = 1 on every curve, as for a generic curve. Why trace-zero b adds a factor 2 for a = 0 is not established here; flagged for follow-up, not a result. Divisibility by 3 in class B ran at 27/63, 126/255, 495/1023, 1269/4095, 8127/16383 for a = 0 (l = 3..7), not compared with a generic arm.

Arm D (b in k*, defined over k; counts of 2^l-1 curves): r_max > sqrt #E(K) for 49 and 42 of 127 curves at l = 7 (a = 0 / Tr(a)=1), 18/30 of 63 at l = 6, and so on (`data/run.log`); small, shown only as the contrast.

Controls and cross-checks

control	result
positive: b in k gives m = 1	all 2^l - 1 such b have m=1, genus 1, every l=2..7

positive: toy census l = 4 reproduced	15 / 255 / 3825
positive: cond (8) detector, n = 2	equals derived counts at l = 3, 4, 5
point-count formula vs brute force	0 mismatches (l = 2, 3, all b, both a)
Hasse, parity, mod 4, twist, Frobenius invariance	0 violations, all class-B curves, l=2..7
cross-check vs `target/release/ghs_screen`	60 rows (l = 3..7, 12 each; classes A, B and random; a = 0 and Tr(a)=1): magic number, genus, GHS type, condition (8) all equal, 0 mismatches (`data/crosscheck_ghs_screen.csv`)
negative: generic b and null integers	see Q3 table

Failures and limits

- No run failed. The brief's #E(F_{2^l}) criterion is ill-posed for genus-3 b (above); this was declared in the protocol before data.
- l = 2, 3 samples are tiny; Wilson intervals at l = 4 on orbit representatives (23 orbits) are wide.
- The primitive polynomial used per N is the smallest found (listed in `data/q1_census.csv`); counts do not depend on it.
- P3 numbers use one random sample (fixed seed) per l; G and I intervals ignore seed variation. The "8 | #E" observation was seen in the order list `data/classB_orders.csv`, post hoc; no hypothesis test was run on it.
- Whether the genus-3 cover actually descends to a smooth model over F_{2^l} and the cost of index calculus on it were not examined (condition (8) is necessary, not sufficient; see `docs/GHS_SCREEN.md`).
- PDF: built from this README with Pandoc (to Typst) and the Typst wheel (`typst 0.15.0`, installed in a scratch virtualenv outside the repository); no system PDF engine was present.

Files

`PROTOCOL.md`, `README.md`, `README.pdf`, `graph.svg`, `graph.png`, `tool/census.rs`, `data/{q1_census,q2_cond8,q3_shaped,classB_orders,crosscheck_ghs_screen}.csv`, `data/run.log`, `data/zscores.txt`, `data/graph.svg` (tool output; `graph.svg` is a copy).

`report.typ` is the Pandoc-generated Typst source of the PDF.